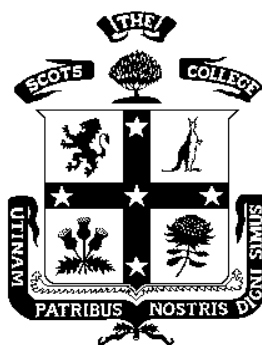

THE SCOTS COLLEGE



MATHEMATICS EXTENSION I

YEAR 12 PRETRIAL

5TH APRIL 2013

GENERAL INSTRUCTIONS

- Reading time – 5 minutes
- Working time - 1.5 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided
- Show all necessary working in Section II

WEIGHTING

30%

TOTAL MARKS

55

SECTION I (7 MARKS)

- Answers to be recorded on the multiple choice answer sheet provided
- Clearly label your answer sheet with your student number
- Allow about 10 minutes for this section

SECTION II (48 MARKS)

- Questions 8 - 11
- Answers to be recorded in the answer booklets provided
- *Each question must be completed in a new answer booklet.*
- Label each answer booklet with your student number and the question number attempted. Clearly indicate the booklet order if more than one booklet is used for a question. Eg) Book 1 of 2 and 2 of 2.

SECTION I

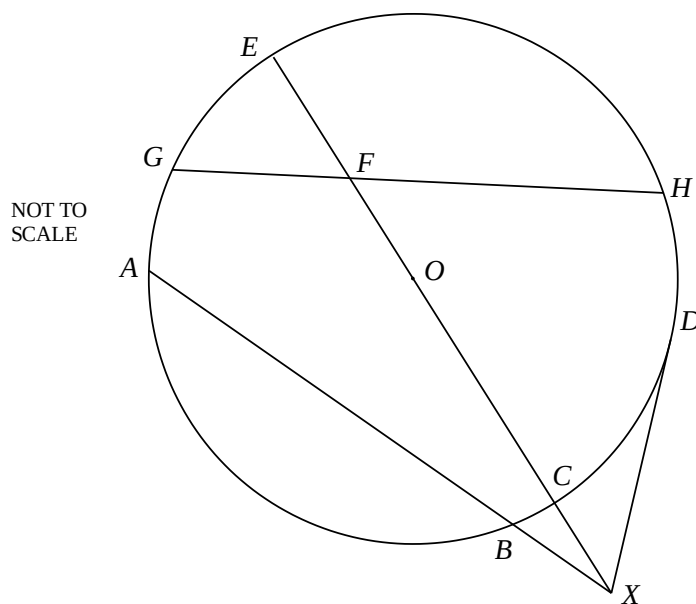
QUESTION 1

Which of the following is the list of all the factors of $P(x) = 3x^3 + 4x^2 - 5x - 2$

- (A) $(x + 1)(x - 2)(x - 3)$
- (B) $(x + 1)(x + 2)(x + 3)$
- (C) $3x(x - 1)(x + 1)$
- (D) $(x - 1)(x + 2)(3x + 1)$

QUESTION 2

A circle of centre O has a radius of 6 cm. From an external point X , a tangent is drawn with a point of contact D . From X the secants XA and XE are also drawn.



If $DX = 8$ cm calculate the distance CX .

- (A) 4 cm
- (B) -16 cm
- (C) 8 cm
- (D) 6 cm

QUESTION 3

The exact value of $\sin 75^\circ$ is:

- (A) $\frac{4-\sqrt{6}}{4}$
- (B) $\frac{2+\sqrt{2}}{2\sqrt{2}-1}$
- (C) $\frac{\sqrt{6}+\sqrt{2}}{4}$
- (D) $\frac{\sqrt{2}-\sqrt{6}}{4}$

QUESTION 4

Evaluate: $\int \frac{dx}{\sqrt{9-2x^2}}$

- (A) $y = \sin^{-1} \sqrt{2}x + c$
- (B) $y = \frac{1}{2} \cos \frac{2x}{3} + c$
- (C) $y = \frac{1}{\sqrt{2}} \sin^{-1} \frac{\sqrt{2}x}{3} + c$
- (D) $y = \frac{1}{\sqrt{2}} \log_e \sqrt{2}x + c$

QUESTION 5

Calculate the acute angle between the lines $l_1: 2y - 3x = 7$ and $l_2: 2x - 5y + 1 = 0$, to the nearest degree.

- (A) 35°
- (B) 70°
- (C) 78°
- (D) 282°

QUESTION 6

Given $\frac{d}{dx} \left(\frac{2x}{4+x^2} + \tan^{-1} \frac{x}{2} \right) = \frac{16}{(4+x^2)^2}$, evaluate: $\int_0^2 \frac{dx}{(4+x^2)^2}$

- (A) $\frac{\pi}{16}$
- (B) $\pi + 4$
- (C) $\frac{\pi+2}{64}$
- (D) $2\pi - 1$

QUESTION 7

For the curve $y = \frac{3x^2+1}{x^2+2x}$, the vertical and horizontal asymptotes are:

- (A) $x = 0, x = -2, y = 3$
- (B) $x = \frac{1}{3}, y = 0, y = -1$
- (C) $x = 0, x = \sqrt{2}, y = 1$
- (D) $x = 6, y = -2, y = -1$

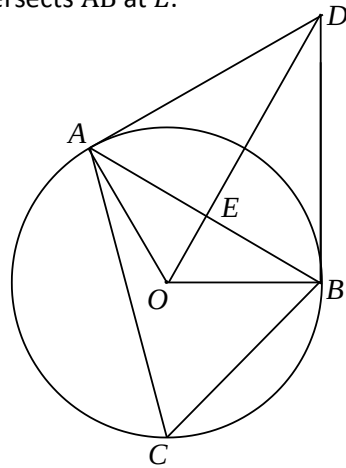
SECTION II

QUESTION 8

(START A NEW ANSWER BOOKLET)

12 MARKS

- a) The diagram below shows the points A , B , and C on a circle with centre O . Tangents are drawn from A and B which meet at D . O is joined to D and the interval OD intersects AB at E .



Not to
Scale

- i) Prove that $\angle AOB = 2 \times \angle DAB$. 2 marks
- ii) Prove that $AOBD$ is a cyclic quadrilateral. 1 mark
- iii) Prove that E is the midpoint of AB . 2 marks
- b) Find the coordinates of the point $P(x, y)$ which divides the interval AB internally in the ratio $4 : 9$ with $A(2, 3)$ and $B(5, -7)$. 2 marks
- c) Solve the inequality $\frac{x(x-3)}{x-2} > 2$. 3 marks
- d) Differentiate: $y = \sin^{-1} \frac{1}{4}(2x - 3)$ 2 marks

END OF QUESTION 8

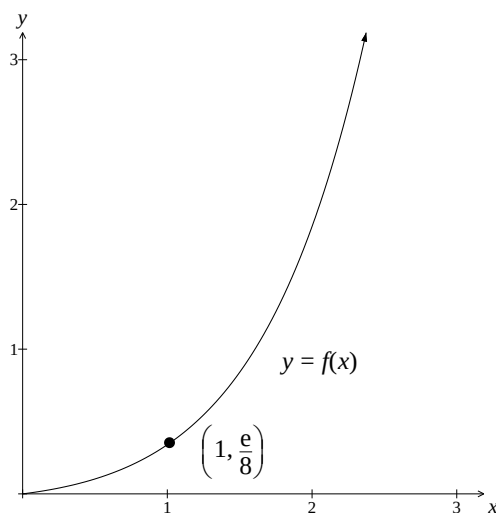
- a)** Given the parametric coordinates: $x = 2t$ and $y = t^2$.
- i)** Show that the Cartesian equation of the parabola is: $x^2 = 4y$ 1 mark
- ii)** Given the parameter $t = 2$, show that the equation of the normal at that point is $x + 2y - 12 = 0$ 2 marks
- iii)** Find the point of intersection of the normal and the x – axis 1 mark
- b)** Given the inverse trigonometric function: $y = 3 \cos^{-1}(2x)$.
- i)** State the domain and range of $y = 3 \cos^{-1}(2x)$ 2 marks
- ii)** Find the gradient function of $y = 3 \cos^{-1}(2x)$ 2 marks
- iii)** Find the equation of the tangent to $y = 3 \cos^{-1}(2x)$ at $x = 0$ 2 marks
- c)** Find the general solution of $\sin \theta = \frac{\sqrt{2}}{2}$ 2 marks

- a) The polynomial $P(x) = x^4 - 3x^3 + ax^2 + bx - 6$ leaves a remainder of 8 when divided by $(x + 1)$. If $x - 3$ is a factor of $P(x)$, find a and b . 3 marks

- b) Evaluate: $\int_0^{\frac{\pi}{4}} \sin^2 x \, dx$ 2 marks

- c) i) Show by differentiation that $y = \frac{xe^x}{2}$ is increasing for $x \geq 0$. 2 marks

- ii) A sketch of $y = f(x) = \frac{xe^x}{2}$; $x \geq 0$ is shown below. Explain why $y = f(x)$ has an inverse function. 1 mark



- iii) Copy the graph above and add a sketch of the inverse function $y = f^{-1}(x)$. 1 mark

- d) i) Write $\sqrt{3} \cos x - \sin x$ in the form $A \cos(x + \phi)$; $0 < \phi < \frac{\pi}{2}$. 2 marks

- ii) Hence, or otherwise solve the equation $\sqrt{3} \cos x - \sin x = 1$; $0 \leq x \leq 2\pi$. 1 mark

- a) Find $\int_1^6 x\sqrt{x+3} \, dx$, given the substitution $u = \sqrt{x+3}$. 4 marks
- b) For the function $y = \frac{\log_e x}{x}$;
- i) State the domain of the function 1 mark
 - ii) Find any stationary points and determine their nature 2 marks
 - iii) Find the x - intercept 1 mark
 - iv) Hence, sketch the function including the above information and showing the property of the curve as $x \rightarrow \infty$ 2 marks
- c) Find the Cartesian equation of the curve represented by the following parametric equations: 2 marks
- $$x = \sin 2t, \quad y = \cos t$$



THE SCOTS COLLEGE – MATHEMATICS 2013
EXTENSION 1 MATHEMATICS PRE-TRIAL HSC

CANDIDATE NUMBER: _____

SECTION I – MULTIPLE CHOICE ANSWER SHEET (7 MARKS)

Mark the correct answer by filling in the circle. To make a correction, neatly place a cross over the circle and then fill in the correct circle.

EXAMPLE:

A	B	C	D
☐	☒	☒	☐

	A	B	C	D
Question 1	☐	☐	☐	☐
Question 2	☐	☐	☐	☐
Question 3	☐	☐	☐	☐
Question 4	☐	☐	☐	☐
Question 5	☐	☐	☐	☐
Question 6	☐	☐	☐	☐
Question 7	☐	☐	☐	☐

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

SECTION 1. MULTIPLE CHOICE.

Summary:

Q1. D

Q2. A

Q3. C

Q4. C

Q5. A

Q6. C

Q7. A

Solutions:

$$Q1. P(x) = 3x^3 + 4x^2 - 5x - 2$$

$$P(1) = 3(1)^3 + 4(1)^2 - 5(1) - 2$$

$$= 0$$

$\therefore (x-1)$ is a factor

$$\begin{array}{r} 3x^2 + 7x + 2 \\ x-1 \overline{) 3x^3 + 4x^2 - 5x - 2} \\ \underline{3x^3 - 3x^2} \\ 7x^2 - 5x \\ \underline{7x^2 - 7x} \\ 2x - 2 \\ \underline{2x - 2} \\ 0 \end{array}$$

$$\begin{aligned} \therefore P(x) &= (x-1)(3x^2 + 7x + 2) \\ &= (x-1)(3x^2 + 6x + x + 2) \\ &= (x-1)[3x(x+2) + 1(x+2)] \\ &= (x-1)(3x+1)(x+2) \end{aligned}$$

(D)

$$Q2. DX^2 = (CX)(XE)$$

ratio of secant = square of tangent

$$8^2 = (CX)(CX + 12)$$

$$64 = CX^2 + 12CX$$

$$CX^2 + 12CX - 64 = 0$$

$$(CX + 16)(CX - 4) = 0$$

$$\therefore CX = -16 \text{ and } CX = 4$$

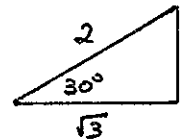
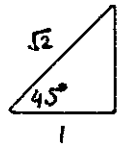
$$CX \neq -ve$$

$$\therefore CX = 4 \text{ cm}$$

(A)

$$Q3. \sin 75^\circ = \sin(45^\circ + 30^\circ)$$

$$= \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ$$



$$\sin 45^\circ = \frac{1}{\sqrt{2}}$$

$$\sin 30^\circ = \frac{1}{2}$$

$$\cos 45^\circ = \frac{1}{\sqrt{2}}$$

$$\cos 30^\circ = \frac{\sqrt{3}}{2}$$

$$\therefore \sin 75^\circ = \left(\frac{1}{\sqrt{2}}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{2}\right)$$

$$= \frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}}$$

$$= \frac{\sqrt{3} + 1}{2\sqrt{2}}$$

$$= \frac{\sqrt{6} + \sqrt{2}}{4}$$

(C)

$$\begin{aligned}
 Q4. \int \frac{dx}{\sqrt{9-2x^2}} &= \int \frac{dx}{\sqrt{2(\frac{9}{2}-x^2)}} \\
 &= \int \frac{dx}{\sqrt{2}\sqrt{(\frac{3}{\sqrt{2}})^2-x^2}} \\
 &= \frac{1}{\sqrt{2}} \int \frac{dx}{\sqrt{(\frac{3}{\sqrt{2}})^2-x^2}} \\
 &= \frac{1}{\sqrt{2}} \sin^{-1} \frac{x}{\frac{3}{\sqrt{2}}} + C \\
 &= \frac{1}{\sqrt{2}} \sin^{-1} \frac{\sqrt{2}x}{3} + C
 \end{aligned}$$

(C)

$$\begin{aligned}
 Q5. l_1: 2y - 3x &= 7 \\
 \therefore y &= \frac{3}{2}x + \frac{7}{2}
 \end{aligned}$$

$$\begin{aligned}
 l_2: 2x - 5y + 1 &= 0 \\
 \therefore y &= \frac{2}{5}x + \frac{1}{5}
 \end{aligned}$$

$$m_1 = \frac{3}{2} \quad m_2 = \frac{2}{5}$$

$$\begin{aligned}
 \therefore \tan \theta &= \frac{\frac{3}{2} - \frac{2}{5}}{1 + (\frac{3}{2})(\frac{2}{5})} \\
 &= \frac{\frac{11}{10}}{\frac{8}{5}}
 \end{aligned}$$

$$\theta = 35^\circ$$

(A)

$$\begin{aligned}
 Q6. \int_0^2 \frac{dx}{(4+x^2)^2} &= \frac{1}{16} \int_0^2 \frac{16}{(4+x^2)^2} dx \\
 &= \frac{1}{16} \left[\frac{2x}{4+x^2} + \tan^{-1} \frac{x}{2} \right] \Big|_0^2 \\
 &= \frac{1}{16} \left[\left(\frac{2(2)}{4+(2)^2} + \tan^{-1} \left(\frac{2}{2} \right) \right) - 0 \right] \\
 &= \frac{1}{16} \left[\frac{4}{8} + \frac{\pi}{4} - 0 \right] \\
 &= \frac{2+\pi}{64}
 \end{aligned}$$

(C)

Q7. Vertical:

$$y = \frac{3x^2+1}{x(x+2)}$$

i.e. $x \neq 0$ and $x \neq -2$.

Horizontal:

$$\begin{aligned}
 \lim_{x \rightarrow \infty} \frac{3x^2+1}{x(x+2)} &= \lim_{x \rightarrow \infty} \frac{\frac{3x^2}{x^2} + \frac{1}{x^2}}{\frac{x^2}{x^2} + \frac{2x}{x^2}} \\
 &= \lim_{x \rightarrow \infty} \frac{3 + \frac{1}{x^2}}{1 + \frac{2}{x}} \\
 &= \frac{3+0}{1+0} \\
 &= 3.
 \end{aligned}$$

$\therefore$ Asymptotes:

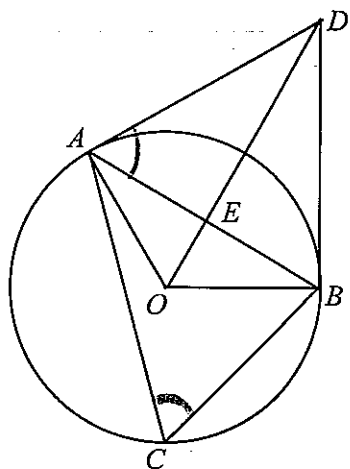
$$x=0, x=-2, y=3$$

(A)

END SECTION I.

SECTION II.

Q8



Not to Scale

ai. Prove $\angle AOB = 2 \times \angle DAB$

$$\angle DAB = \angle ACB$$

($\angle$ between tangent and chord
= $\angle$ in alternate segment)

$$\angle AOB = 2 \times \angle ACB$$

($\angle$ at centre is twice the $\angle$
at circumference on same arc)

$$\therefore \angle AOB = 2 \times \angle DAB$$

(since $\angle DAB = \angle ACB$)

* 2 marks for a complete proof

* 1 mark if some relevant facts
are stated or proof is incom-
plete.

a.ii. Prove that AOB is a cyclic
quadrilateral.

$$\angle DAO = \angle DBO = 90^\circ$$

(tangent perpendicular to radius)

$$\therefore \angle DAO + \angle DBO = 180^\circ$$

(sum of two right angles)

$\therefore$ Opposite angles of AOB are
supplementary.

$\therefore$ AOB is a cyclic quadrilateral

* 1 mark as long as statement
that opposite angle are supplem-
entary and why.

a.iii. Prove: E is the midpoint of
AB.

$$AO = BO \quad (\text{equal radii})$$

$$AD = BD \quad (\text{tangents from
an external point are equal in length})$$

$\therefore$ AOB is a kite

$\therefore$ OD bisects AB (symmetry
of a kite)

$\therefore$ E is midpoint of AB.

$$8b. A(2,3), B(5,-7) \quad \frac{m}{n} = \frac{4}{9}$$

$$x = \frac{mx_2 + nx_1}{m+n}$$

$$= \frac{20+18}{13}$$

$$= \frac{38}{13}$$

$$y = \frac{my_2 + ny_1}{m+n}$$

$$= \frac{-28+27}{13}$$

$$= -\frac{1}{13}$$

$\therefore$ P is $(\frac{38}{13}, -\frac{1}{13})$

* 2 marks for correct coordinates

* 1 mark for one of the coordinates
correct or for a satisfactory
amount of working correct
and shown.

$$8c. \frac{x(x-3)}{x-2} > 2$$

$$\frac{x^2-3x}{x-2} > 2$$

$$\frac{x^2-3x-2}{x-2} > 0, x \neq 2$$

$$\frac{x^2-3x}{x-2} - \frac{2x-4}{x-2} > 0$$

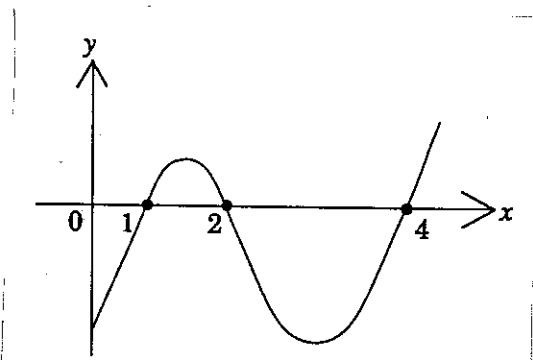
$$\frac{x^2-5x+4}{x-2} > 0$$

Multiply by $(x-2)^2$ (always > 0)

$$\therefore (x-4)(x-1)(x-2) > 0$$

$$\therefore y = (x-1)(x-2)(x-4)$$

$$y = 0 \text{ when } x = 1, 2, 4$$



$$\therefore \frac{x(x-3)}{x-2} > 2 \text{ for } 1 < x < 2 \text{ and } x > 4$$

$$8d. \frac{d}{dx} \sin^{-1} \frac{1}{4}(2x-3) = \frac{d}{dx} \sin^{-1} \left(\frac{1}{2}x - \frac{3}{4} \right)$$

$$= \frac{\frac{1}{2}}{\sqrt{1 - \left(\frac{1}{2}x - \frac{3}{4} \right)^2}}$$

$$= \frac{\frac{1}{2}}{\sqrt{1 - \left(\frac{1}{2}x - \frac{3}{4} \right)^2}} \times \frac{\sqrt{4}}{\sqrt{4}}$$

$$= \frac{1}{\sqrt{4 - 4\left(\frac{x^2}{4} - 3x + \frac{9}{16}\right)}}$$

$$= \frac{1}{\sqrt{4 - x^2 + 3x - \frac{9}{4}}}$$

$$= \frac{1}{\sqrt{\frac{7}{4} + 3x - x^2}}$$

we don't like fraction in the denominator.

$$= \frac{\sqrt{4}}{\sqrt{4}} \times \frac{1}{\sqrt{\frac{7}{4} + 3x - x^2}}$$

$$= \frac{2}{\sqrt{7 + 12x - 4x^2}}$$

$$9a. x = 2t \quad y = t^2$$

$$ai. x = 2t$$

$$x^2 = 2t^2$$

$$\therefore x^2 = 4y$$

$$aii. t = 2, y = \frac{x^2}{4} \quad P(4, 4)$$

$$\therefore y' = m = \frac{2x}{4} = \frac{x}{2} = \frac{2t}{2} = t$$

$$\therefore m = 2 \rightarrow \text{tangent}$$

$$\therefore \text{Normal } m = -\frac{1}{2}$$

$$y - y_1 = m(x - x_1)$$

$$y - 4 = -\frac{1}{2}(x - 4)$$

$$-2y + 8 = x - 4$$

$$-x - 2y + 12 = 0$$

$$x + 2y - 12 = 0 \text{ as required.}$$

$$t = 2 \quad x + 2y = 2(1)(2) + (1)(2)^3$$

$$a = 1. \quad x + 2y = 12$$

$$x + 2y - 12 = 0 \text{ as required.}$$

$$aiii. \quad y = 0$$

$$\therefore x + 0 - 12 = 0$$

$$x = 12$$

$\therefore$ Point of intersection : (12, 0)

9b. $y = 3 \cos^{-1}(2x)$

bi. D: $\cos^{-1}x \quad -1 \leq x \leq 1$

D: $\cos^{-1}2x \quad -1 \leq 2x \leq 1$
 $-\frac{1}{2} \leq x \leq \frac{1}{2}$

R: $\cos^{-1}x \quad 0 \leq x \leq \pi$

R: $3 \cos^{-1}(2x) \quad 0 \leq x \leq 3\pi$

bii. $y = 3u$ where $u = \cos^{-1}(2x)$

$$\begin{aligned} \therefore \frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} \\ &= 3 \times \frac{-2}{\sqrt{1-(2x)^2}} \\ &= \frac{-6}{\sqrt{1-4x^2}} \end{aligned}$$

biii. when $x=0$, $y = 3 \cos^{-1}(2 \times 0)$

$$y = 3 \times \frac{\pi}{2}$$

$$y = \frac{3\pi}{2}$$

$$\begin{aligned} m = y' &= \frac{-6}{\sqrt{1-0}} \\ &= -6 \end{aligned}$$

$$\therefore y - y_1 = m(x - x_1)$$

$$y - \frac{3\pi}{2} = -6x$$

$$6x + y - \frac{3\pi}{2} = 0$$

$$6x + y - \frac{3\pi}{2} = 0$$

9c. $x = \sin^{-1}a + 2n\pi$

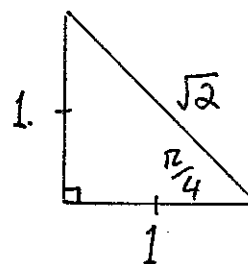
or.

$$x = (\pi - \sin^{-1}a) + 2n\pi$$

where n is an integer.

If $x = \sin^{-1} \frac{\sqrt{2}}{2}$ then

$$x = \sin^{-1} \frac{\sqrt{2}}{2} + 2n\pi \text{ or } x = (\pi - \sin^{-1} \frac{\sqrt{2}}{2}) + 2n\pi$$



$$\therefore \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

$$\therefore \sin^{-1} \frac{\sqrt{2}}{2} = \frac{\pi}{4}$$

Hence.

$$x = \frac{\pi}{4} + 2n\pi \text{ or } x = \frac{3\pi}{4} + 2n\pi$$

10a. $P(-1) = 8$

$$\therefore (-1)^4 - 3(-1)^3 + a(-1)^2 + b(-1) - 6 = 8$$

$$1 + 3 + a - b - 6 = 8$$

$$a - b - 2 = 8$$

$$a - b = 10$$

$$P(3) = 0$$

$$3^4 - 3(3)^3 + a(3)^2 + b(3) - 6 = 0$$

$$81 - 81 + 9a + 3b - 6 = 0$$

$$9a + 3b = 6$$

$$3a + b = 2$$

$$\textcircled{1} \dots a - b = 10$$

$$\textcircled{2} \dots 3a + b = 2$$

$$\textcircled{1} + \textcircled{2}$$

$$4a = 12$$

$$a = 3$$

sub $a=3$ into $\textcircled{1}$

$$3 - b = 10$$

$$-b = 7$$

$$b = -7$$

$$10b. \int_0^{\pi/4} \sin^2 x \, dx$$

$$\text{Now } \cos 2x = 1 - 2\sin^2 x$$

$$\therefore 2\sin^2 x = 1 - \cos 2x$$

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

$$\begin{aligned} \therefore \int_0^{\pi/4} \sin^2 x \, dx &= \\ &= \int_0^{\pi/4} \frac{1}{2}(1 - \cos 2x) \, dx \\ &= \frac{1}{2} \int_0^{\pi/4} (1 - \cos 2x) \, dx \\ &= \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\pi/4} \\ &= \frac{1}{2} \left[\frac{\pi}{4} - \frac{1}{2} \sin \frac{\pi}{2} - 0 \right] \\ &= \frac{\pi}{8} - \frac{1}{4} \sin \frac{\pi}{2} = \frac{\pi}{8} - \frac{1}{4} \end{aligned}$$

$$10ci. \, y = \frac{xe^x}{2}$$

$$y' = \frac{x}{2} \times \frac{d(e^x)}{dx} + e^x \times \frac{d(\frac{x}{2})}{dx}$$

$$y' = \frac{x}{2} \times e^x + e^x \times \frac{1}{2}$$

$$y' = \frac{xe^x + e^x}{2}$$

For $x > 0$, $e^x > 1$

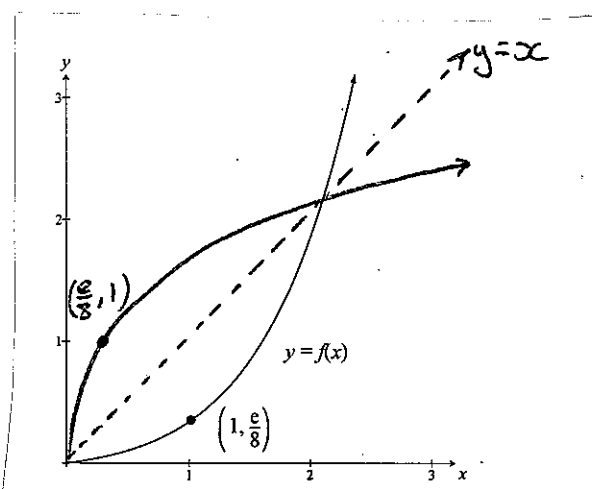
$\therefore xe^x > 0$ (positive $\times$ positive)

$\therefore xe^x + e^x > 0$ (positive + positive)

$\therefore \frac{xe^x + e^x}{2} > 0$ (positive $\div$ positive)

10cii. For a given function $f(x)$, it must be one-to-one or pass a horizontal line test for the inverse of the function $f^{-1}(x)$ to likewise be one-to-one and pass a vertical line test in the given domain.

10ciii.



* As long as the graph appears to be a reflection of the original curve in $y=x$, mark awarded.

10 di. Let $\sqrt{3} \cos x - \sin x = A \cos(x + \phi)$

$\therefore \sqrt{3} \cos x - \sin x = A \cos x \cos \phi - A \sin x \sin \phi$

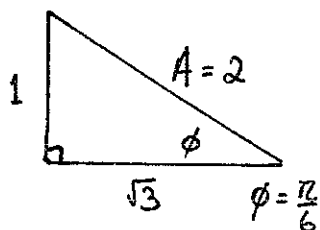
$\therefore \sqrt{3} \cos x = A \cos x \cos \phi \quad - \sin x = A \sin x \sin \phi$

$\sqrt{3} = A \cos \phi$

$1 = A \sin \phi$

$\cos \phi = \frac{\sqrt{3}}{A}$

$\sin \phi = \frac{1}{A}$



$\therefore \sqrt{3} \cos x - \sin x = 2 \cos(x + \frac{\pi}{6})$

dii $\sqrt{3} \cos x - \sin x = 1$

$\therefore 2 \cos(x + \frac{\pi}{6}) = 1$ from part i.

$\cos(x + \frac{\pi}{6}) = \frac{1}{2}$

$\therefore x + \frac{\pi}{6} = \frac{\pi}{3}$ or $(2\pi - \frac{\pi}{3})$

$\therefore x = \frac{\pi}{6}$ or $x = \frac{3\pi}{2}$

11a. $\int_1^6 x/\sqrt{x+3} dx$ given

$u = \sqrt{x+3} \quad \frac{du}{dx} = \frac{1}{2\sqrt{x+3}}$

$\therefore u^2 = x+3$

$\therefore x = u^2 - 3 = \frac{1}{2u}$

$\therefore dx = 2u du$

when $x = 1$

$u = \sqrt{1+3} = 2$

$x = 6$

$u = \sqrt{6+3} = 3$

$\therefore \int_1^6 x/\sqrt{x+3} dx =$

$= \int_2^3 (u^2 - 3)(u)(2u) du$

$= 2 \int_2^3 (u^2 - 3)(u^2) du$

$= 2 \int_2^3 u^4 - 3u^2 du$

$= 2 \left[\frac{u^5}{5} - u^3 \right]_2^3$

$= 2 \left[\frac{243}{5} - 27 - \left(\frac{32}{5} - 8 \right) \right]$

$= 46 \frac{2}{5}$

11b. $x = \sin 2t, y = \cos t$

$\therefore x = 2 \sin t \cos t$

$\therefore x^2 = 4 \sin^2 t \cos^2 t$

$x^2 = 4 \cos^2 t (1 - \cos^2 t)$

$x^2 = 4 y^2 (1 - y^2)$

11ci. $x \in \mathbb{R}$ where $x > 0$.

cii. $y = \frac{\log_e x}{x}$

$y' = \frac{x \times \frac{1}{x} - \log_e x \times 1}{x^2}$

$y' = \frac{1 - \log_e x}{x^2}$

stationary points at $y' = 0$

$\frac{1 - \log_e x}{x^2} = 0$

$$\therefore 1 - \log_e x = 0$$

$$\log_e x = 1$$

$$\therefore x = e$$

when $x = e$, $y = \frac{\log_e e}{e}$

$$y = \frac{1}{e}$$

$\therefore$ Stationary point : $(e, \frac{1}{e})$

x	2	e	3
y'	+ve	0	-ve
Slope	/	-	\

NATURE:

$\therefore$ MAXIMUM at $(e, \frac{1}{e})$

ciii. x -intercept at $y = 0$

$$\frac{\log_e x}{x} = 0$$

$$\log_e x = 0$$

$$x = e^0$$

$$x = 1.$$

civ.

